PROJECT REPORT

Hospital Operations Analysis and Resource Optimization Using Power BI

Healthcare analytics and business intelligence

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Project title	Hospital Operations Analysis and Resource Optimization Using Power BI
Programme	B.Tech Computer Science Engineering
Institute	Poornima Institute of Engineering and Technology, Jaipur
Tools used	Power BI Desktop, Power Query (M), DAX, Microsoft Excel, Python
Data period	1 January 2024 to 30 September 2026 (synthetic dataset)
Submitted	October 2026

Contents: 1 Dataset description | 2 Objectives | 3 Data cleaning performed | 4 Data model | 5 DAX measures | 6 Dashboard screenshots | 7 Key findings | 8 Conclusion

1. Dataset description

The project uses a synthetic hospital operations workbook, *Hospital_Operations_Advanced_PowerBI.xlsx*, with 11 sheets covering 1 January 2024 to 30 September 2026 (33 months). It describes 8 departments (Cardiology, Orthopedics, Neurology, General Medicine, Pediatrics, Oncology, Emergency and Maternity), each with one ward, 280 beds, 91 doctors and 1,800 patients. Because the data are synthetic, the findings demonstrate the method and are not real clinical conclusions.

Sheet	Rows	One row represents	Main columns
Departments	8	a department	DepartmentID, DepartmentName, DepartmentGroup
Wards	8	a ward	WardID, WardName, BedCapacity, DepartmentID
Beds	280	a physical bed	BedID, WardID, BedType
Patients	1,800	a patient	PatientID, Age, Gender, City, InsuranceType, AgeGroup
Doctors	91	a doctor	DoctorID, DepartmentID, Specialization, ExperienceYears, Shift
Admissions	3,000	an admission	AdmissionID, PatientID, AdmissionDate, AdmissionType, DoctorID, WardID, BedID, DischargeDate, DischargeStatus, LengthOfStayDays
Discharges	2,864	a completed admission (copy)	DischargeRecordID, AdmissionID, DischargeDate, DischargeStatus
BedSnapshots	8,032	a ward on one day	SnapshotDate, WardID, BedCapacity, OccupiedBeds, AvailableBeds, OccupancyRate
Staffing	8,032	a department on one day	StaffingDate, DepartmentID, DoctorsOnDuty, DoctorsAvailable, NurseOnDuty
Calendar	1,004	a day	Date, Year, Quarter, MonthNumber, MonthName, YearMonth, DayName, IsWeekend
ReadMe	9 items	documentation	Not loaded into the model

2. Objectives

The aim is to give hospital administrators one reliable view of admissions, discharges, bed occupancy, staffing and length of stay, with early warning of capacity pressure. The specific objectives were:

- Audit the workbook and document its data-quality issues before any analysis.
- Clean and shape the data in Power Query without silently deleting incomplete records.
- Build a star-schema model that avoids double counting across fact tables of different grain.
- Create a documented DAX measure library for admissions, beds, staffing, length of stay, time intelligence and alerts.
- Design a five-page interactive dashboard with slicers, drill-through, tooltips and conditional formatting.
- Provide configurable alerts for occupancy, bed availability, admission spikes and staffing load.
- Evaluate forecasts honestly using a holdout test and baselines.
- Validate the headline numbers against the source data.

3. Data cleaning performed

The source workbook was never edited. All cleaning was done as repeatable Power Query steps (3 parameters, 3 helper functions and 11 table queries) after a full audit of keys, missing values, duplicates, orphan IDs, dates and logic errors.

Steps carried out

- Assigned correct data types to every column (IDs as text, dates as date, counts as whole numbers).
- Trimmed spaces and removed non-printing characters from text columns.
- Confirmed there are no duplicate IDs, no orphan foreign keys, and that every date is covered by the Calendar table.
- Added open-stay classification (Completed, Current, Stale open record) using an adjustable 30-day parameter.
- Added data-quality flag columns for overlapping stays and impossible sequences, instead of deleting those records.
- Added check columns confirming that occupied plus available beds equal capacity, and that on-duty doctors never exceed those available.
- Created a Doctors experience band and extra Calendar columns (MonthStart, weekday number, complete-month flag).

- Built a row-count and error check query so each load can be reconciled with the source.

Issues found and how they were handled

Issue	Evidence	Treatment
Missing discharge dates	136 admissions (4.5%), all marked Ongoing; 120 open for more than 30 days	Kept and classified. Excluded from length of stay. The 30-day rule is an assumption.
Duplicate source	Discharges sheet equals the 2,864 completed admissions	Loaded hidden and disconnected; used only for validation.
Snapshots do not match admissions	About 216 occupied beds a day in snapshots, against about 10 inpatients implied by admissions	Occupancy taken only from BedSnapshots.
Overlapping or illogical records	78 bed overlaps, 19 same-patient overlaps, 17 admissions after a Deceased discharge, 66 admissions while an earlier stay is still open	Flagged with DQ columns (177 admissions have at least one flag); not deleted.
Limited staffing data	DoctorsAvailable is constant per department; no nurse roster; no weekday variation	Used as headcount and load context, not performance.
Repeated doctor names	91 doctors but 67 distinct names	Relationships use DoctorID only.

4. Data model

The model is a star schema with three fact tables of different grain and six dimension tables. Facts meet only through shared dimensions, so admissions, bed days and staffing days are never double counted. All relationships are one-to-many with single-direction filtering.

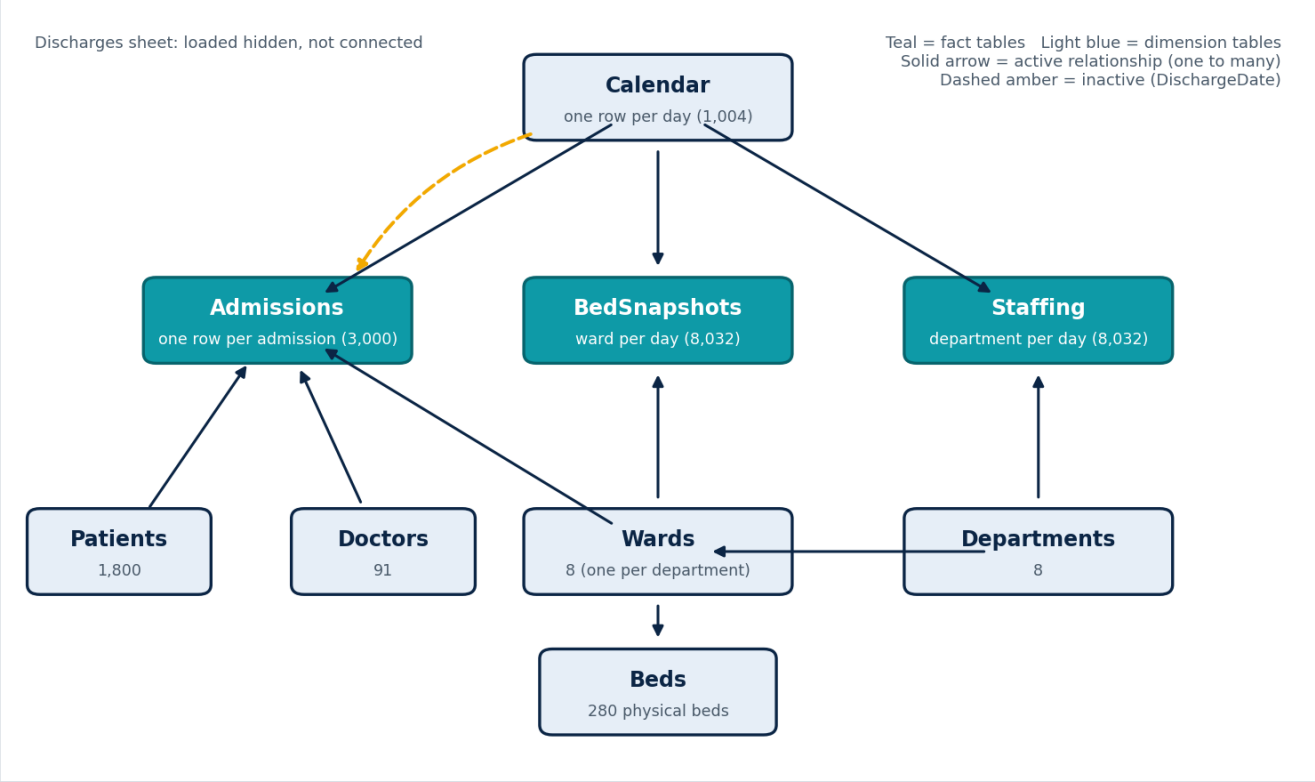


Figure 1. Star-schema data model (generated from the model design).

From (one side)	To (many side)	Notes
Calendar[Date]	Admissions[AdmissionDate]	Active. Default time view for admissions.
Calendar[Date]	Admissions[DischargeDate]	Inactive. Switched on with USERELATIONSHIP for discharge counts.
Calendar[Date]	BedSnapshots[SnapshotDate], Staffing[StaffingDate]	Active. Daily occupancy and staffing.
Departments[DepartmentID]	Wards[DepartmentID], Staffing[DepartmentID]	Passes department filters to wards and staffing.

From (one side)	To (many side)	Notes
Wards[WardID]	Beds, BedSnapshots, Admissions (WardID)	Ward-level beds, occupancy and admissions.
Doctors[DoctorID]	Admissions[DoctorID]	Doctor counts follow the Department slicer through TREATAS.
Patients[PatientID]	Admissions[PatientID]	Demographics and unique patients.

Redundant department links were left out to avoid ambiguous filter paths. The Discharges sheet is hidden and unrelated.

5. DAX measures

A dedicated measures table holds **94 documented measures**, plus 7 validation measures. Alert limits are parameters that the user can change, and are examples, not medical standards.

Folder	Measures	Examples
Parameters	7	Long Stay Days, Warning Occupancy, Staffing Pressure Limit
Admissions and patients	15	Total Admissions, Total Discharges, Unique Patients, Readmission Rate (30-Day)
Beds	24	Average Occupancy Rate, Latest Occupied Beds, High-Occupancy Ward Count (Latest)
Doctors and staffing	13	Total Doctors, Avg Doctors on Duty, Occupied Beds per Doctor on Duty
Length of stay	7	Average, Median, Long-Stay Percentage
Time intelligence	12	Current Month Admissions, MoM and YoY Growth, Rolling 30-Day, YTD
Alerts	11	Admissions Spike Z-Score, Net Admissions, Active Alert Signals (Latest)
Colours	5	Occupancy Color, Spike Color (for conditional formatting)

Selected measures

Measure	Formula	Result
Total Admissions	`COUNTROWS ( Admissions )`	3,000
Total Discharges	`CALCULATE ( COUNTROWS ( Admissions ), USERELATIONSHIP ( Calendar[Date], Admissions[DischargeDate] ), NOT ISBLANK ( Admissions[DischargeDate] ) )`	2,864
Average Occupancy Rate	`DIVIDE ( [Occupied Bed-Days], [Capacity Bed-Days] )`	77.24%
Latest Occupied Beds	`VAR d = [Latest Snapshot Date] RETURN CALCULATE ( SUM ( BedSnapshots[OccupiedBeds] ), BedSnapshots[SnapshotDate] = d )`	198 (30 Sep 2026)
Total Doctors	`CALCULATE ( DISTINCTCOUNT ( Doctors[DoctorID] ), TREATAS ( VALUES ( Departments[DepartmentID] ), Doctors[DepartmentID] ) )`	91
Occupied Beds per Doctor on Duty	`DIVIDE ( [Occupied Bed-Days], SUM ( Staffing[DoctorsOnDuty] ) )`	4.41
Average Length of Stay	`AVERAGE ( Admissions[LengthOfStayDays] )`	3.33 days
Long-Stay Percentage	`DIVIDE ( [Long-Stay Discharge Count], [Completed Stays] )`	11.9%
Admissions MoM Growth %	`DIVIDE ( [Current Month Admissions] - [Previous Month Admissions], [Previous Month Admissions] )`	-22.86% (Sep vs Aug 2026)
Admissions Spike Z-Score	`DIVIDE ( [Total Admissions] - [Monthly Admissions Mean], [Monthly Admissions StdDev] )`	2.03 for Jan 2025

Occupancy is computed as occupied bed-days divided by capacity bed-days, not as an average of daily percentages (77.24% versus a misleading 76.7%).

6. Dashboard screenshots

The report has five pages sharing one design: a navy header, synced Date and Department slicers, six KPI cards, and two rows of charts.

Page 1 - Executive Overview

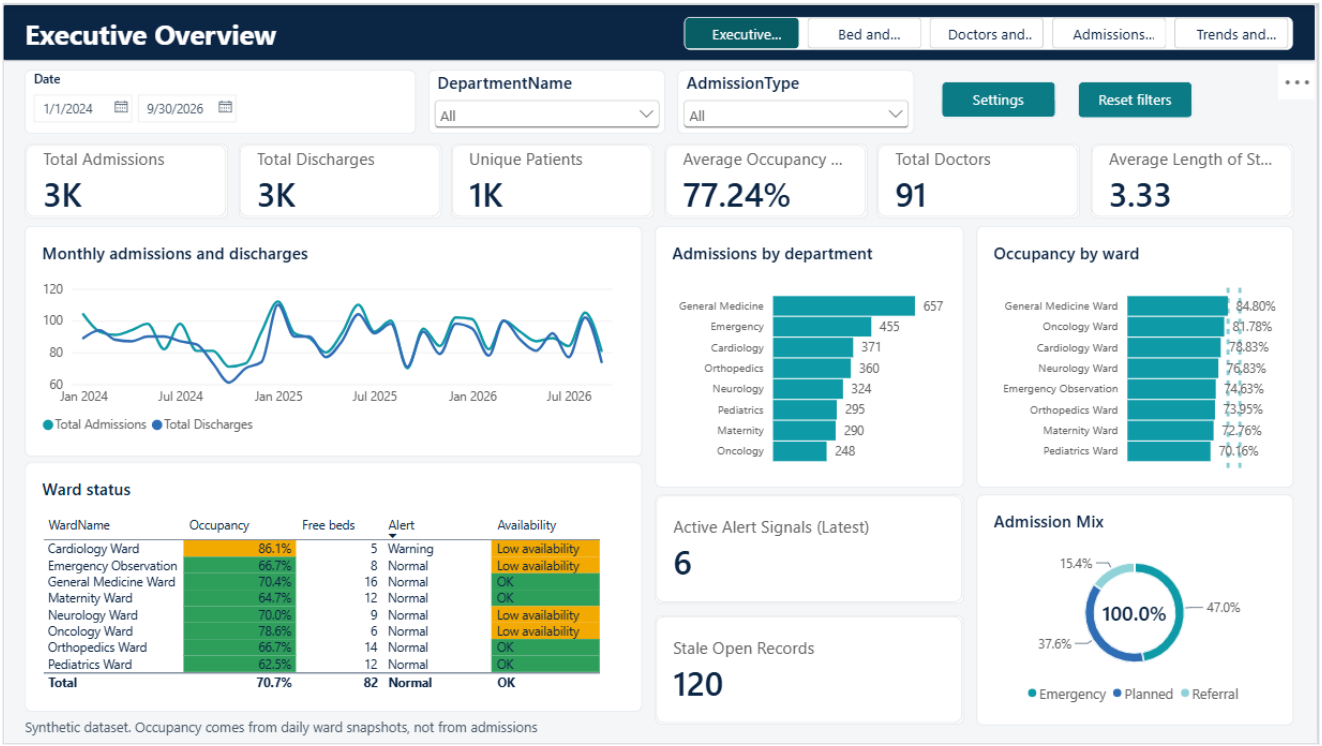


Figure 2. Page 1 - Executive Overview. KPI cards, monthly admissions and discharges, admissions by department, ward occupancy, alert table.

Page 2 - Bed and Capacity Management

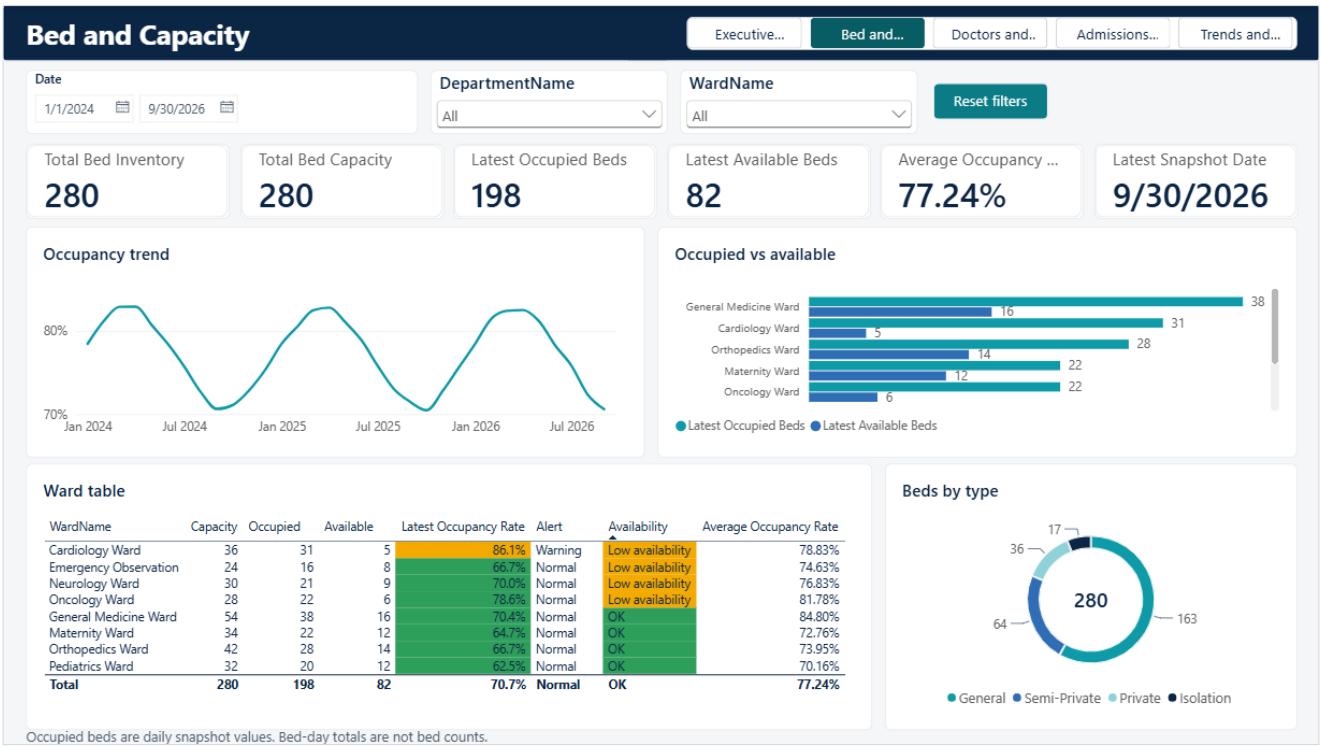


Figure 3. Page 2 - Bed and Capacity Management. Bed inventory, latest occupied and available beds, occupancy trend, ward alert table with drill-through.

Page 3 - Doctors and Staffing

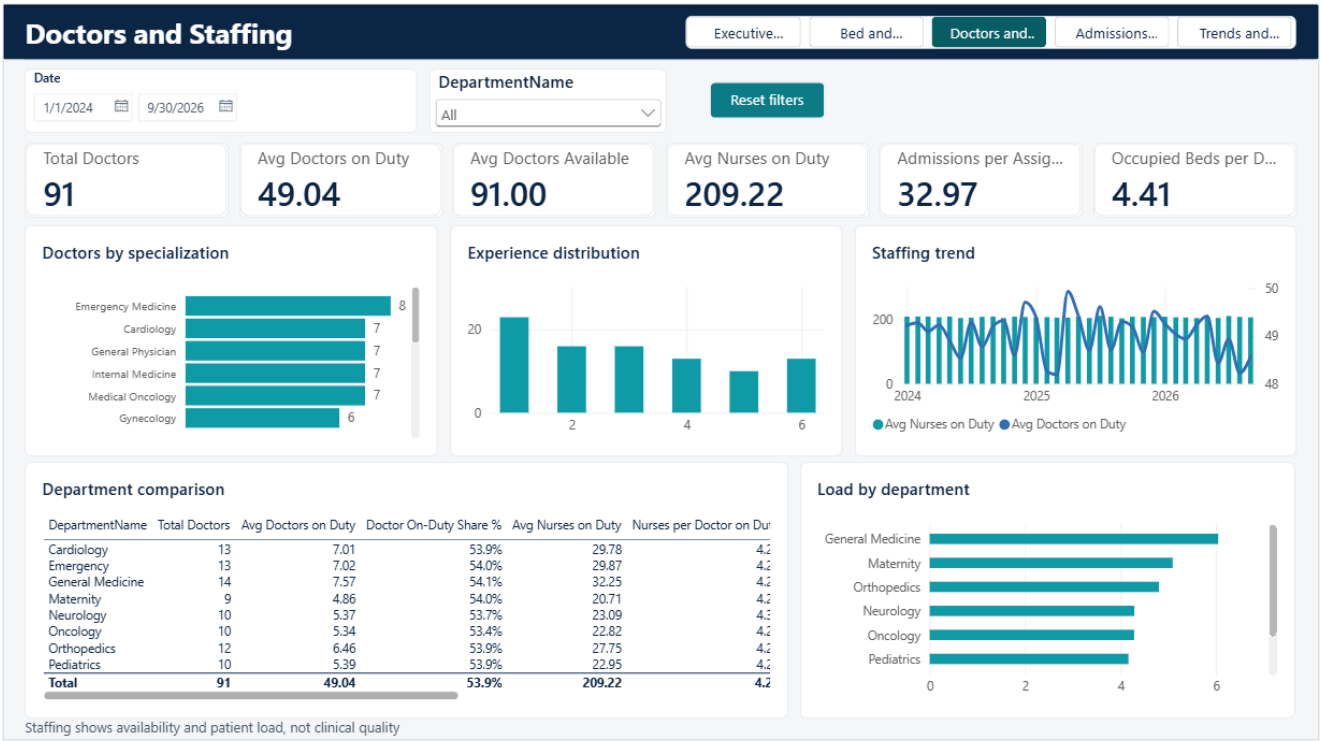


Figure 4. Page 3 - Doctors and Staffing. Doctors by specialization and experience, staffing trend, department comparison and patient load.

Page 4 - Admissions and Discharges

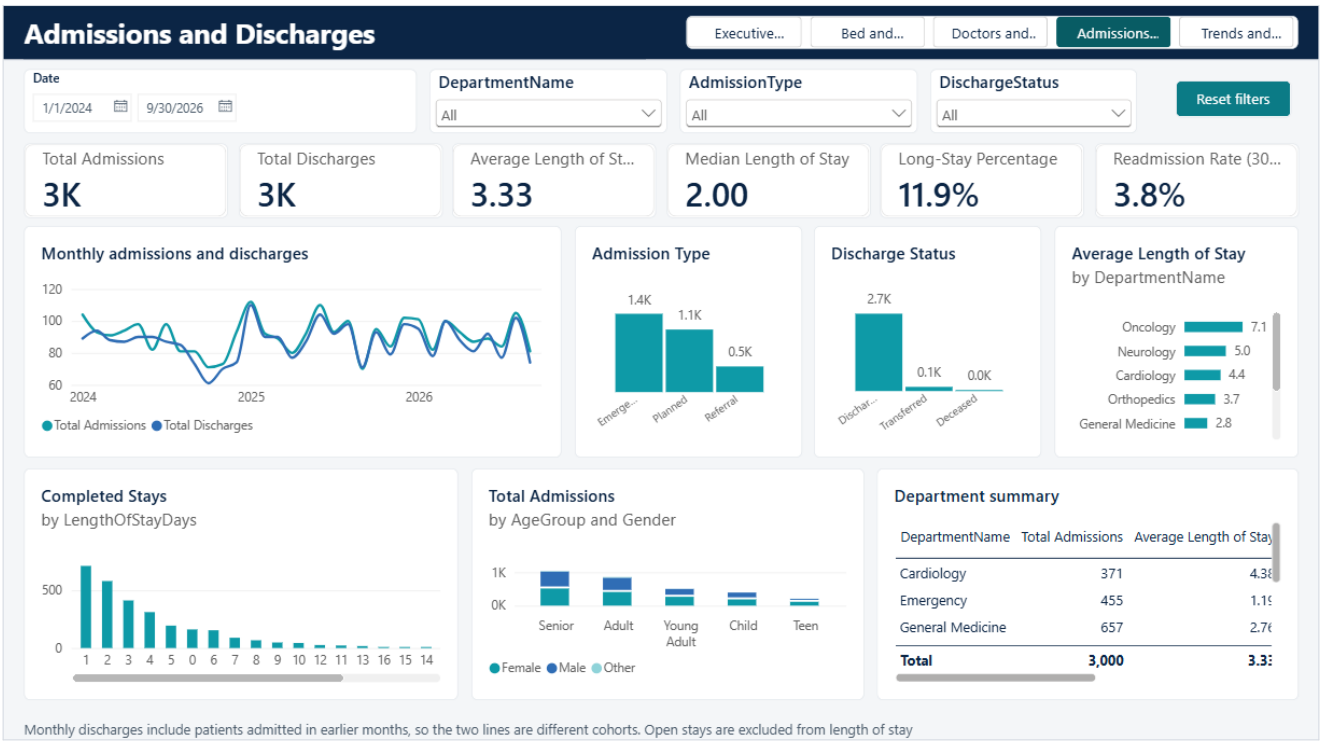


Figure 5. Page 4 - Admissions and Discharges. Admission types, discharge status, length of stay, long stays, readmissions and demographics.

Page 5 - Trends, Forecasting and Insights

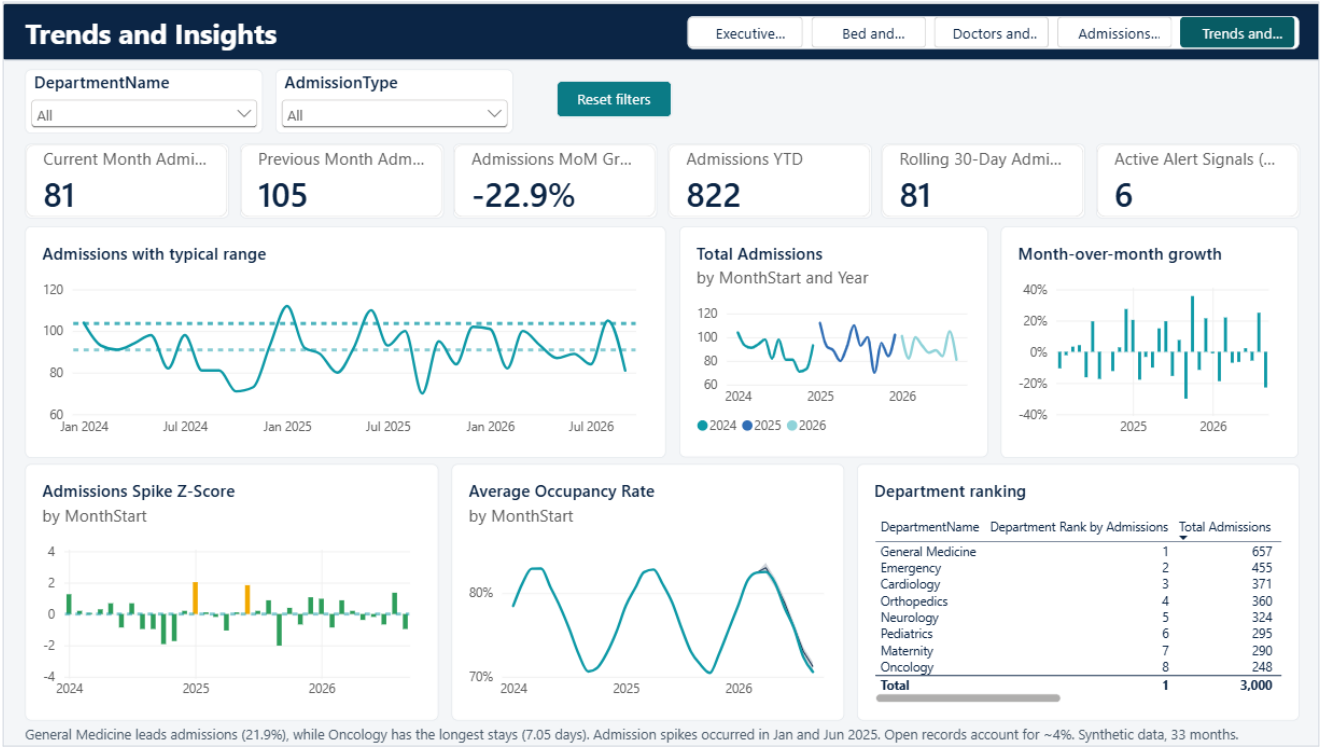


Figure 6. Page 5 - Trends, Forecasting and Insights. Month-over-month change, admission spikes, occupancy forecast and department ranking.

7. Key findings

All figures are for the full period and all departments.

Area	Finding
Volume	3,000 admissions from 1,471 different patients; 2,864 discharges. Emergency 47.0%, Planned 37.6%, Referral 15.4%.
Departments	General Medicine has the largest share of admissions (657, 21.9%); Oncology the smallest (248, 8.3%).
Trend	Admissions averaged 90.9 per month: 1,059 in 2024, 1,119 in 2025 (+5.7%), and 822 in January to September 2026 against 838 in the same months of 2025 (-1.9%). January 2025 (112) and June 2025 (110) were the only spikes.
Length of stay	Mean 3.33 days, median 2, maximum 24. 11.9% of completed stays last 7 days or more. Oncology averages 7.05 days; Emergency 1.19 days.
Occupancy	Overall 77.24%. By ward from 70.2% (Pediatrics) to 84.8% (General Medicine). On 30 Sep 2026, 198 of 280 beds were occupied (70.7%), and the Cardiology ward stood at 86.1%, above the 85% example warning level.
Seasonality	Occupancy peaks in March and April (about 83%) and is lowest in September and October (about 71%). The cycle repeats every year.
Staffing	About 49 of 91 doctors are on duty on an average day. There are 4.41 occupied beds per doctor on duty, highest in General Medicine (6.05) and lowest in Emergency (2.55).
Readmissions	114 thirty-day readmissions, 3.8% of admissions.
Data quality	136 admissions have no discharge date; 120 of them are older than 30 days and are probably missing discharge records.
Forecasting	On a 6-month holdout, a seasonal model forecast occupancy with an average error of 0.27 percentage points against 3.86 for a flat average. Admissions showed no trend or reliable seasonality, and the plain average (about 91 a month, typical range 70 to 112) beat every model, so no admissions forecast is claimed.

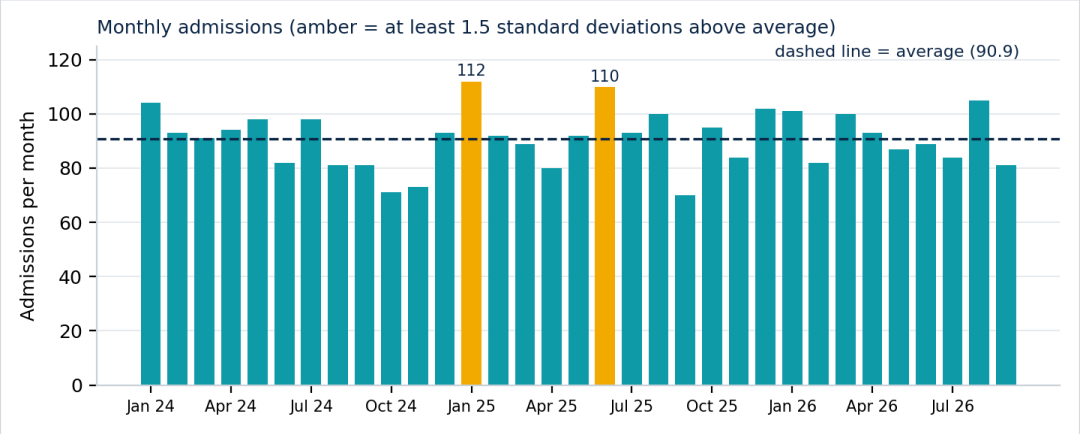


Figure 7. Monthly admissions fluctuate around a flat average. Source: Admissions sheet.

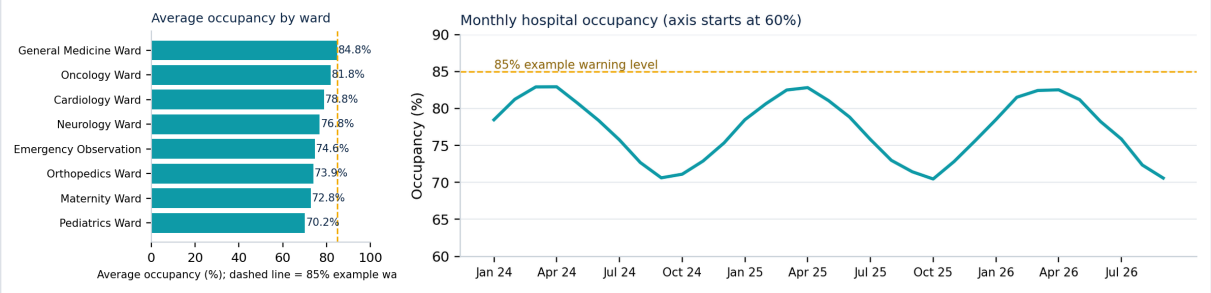


Figure 8. Ward occupancy (left) and monthly hospital occupancy (right). Source: BedSnapshots sheet.

8. Conclusion

The project turned an 11-sheet workbook into a validated, interactive Power BI dashboard. The data were audited and cleaned without deleting incomplete records, modelled as a star schema that prevents double counting, and analysed with 94 documented DAX measures across five report pages with configurable alerts. A self-testing validation page compares the report against the source data.

The analysis shows a hospital with stable admissions of about 91 a month, bed occupancy that follows a regular yearly cycle between about 71% and 83%, and uneven pressure across departments: General Medicine combines the highest ward occupancy with the highest patient load per doctor, and Oncology has the longest stays. Occupancy could be forecast accurately because it is seasonal, whereas admissions behave like random variation around an average, so the honest result is a typical range, not a forecast.

Recommendations (decision support, using example thresholds)

- Ask the records team to review the 120 stale open admissions and enter missing discharge dates.
- Plan elective work around the occupancy cycle, expecting a peak near 83% in March and April and a low near 71% in September and October.
- Monitor the General Medicine and Cardiology wards against the warning level, and review rostering in General Medicine. Staffing load is a workload signal, not a performance judgement.
- Plan admissions with the typical monthly range (70 to 112) and avoid reacting to single-month swings.

Limitations

- The dataset is synthetic with 33 months of history, so the forecasts are an academic prototype and the very regular occupancy cycle is unlikely in real data.
- Bed snapshots and admissions do not reconcile, so bed demand cannot be derived from admissions.
- The 30-day rule for stale open records is an assumption, and alert thresholds are configurable examples.
- Staffing data shows availability and load, not clinical quality.

Future work

- Use real timestamps and patient-level bed stays so occupancy is tied to admissions.
- Add nurse rosters, more years of history, row-level security and scheduled data refresh.